

OpenFOAM v2312 · known-answer verification · VOF / RANS / overset

Computational fluid dynamics (CFD)

Open-source CFD (OpenFOAM) verified against classical analytical and benchmark solutions — known-answer verification across steady-laminar, transient-laminar, turbulent (RANS), free-surface (VOF), wave-generation, 6-DOF floating-body and wave-body interaction (overset) regimes, computed on real OpenFOAM v2312, not mocked. Reports are interactive: live charts, scrubbable animations and a member VIV lock-in screening calculator.

9

verification cases + RAO spot-check

73

real 2D sloshing cases

1.7%

heave RAO vs long-wave limit

WHAT YOU GET

- Free-surface (VOF) canon: dam break (Martin & Moyce), numerical wave tank, green-water impact (Kleefsman / MARIN) and forced-roll tank sloshing within 0.29% of the analytical tanh dispersion
- Overset wave-body interaction: floating-box heave RAO within 1.7% of the long-wave limit, frequency-swept against a frozen potential-flow (BEM) reference
- Cylinder wave loading vs the exact MacCamy-Fuchs diffraction closed form (3.4% at $ka = 0.96$) — a case a Morison estimate fails
- Laminar / turbulent benchmarks: Blasius flat plate, $Re=100$ Karman street (Strouhal within 0.9%), k -omega SST law-of-the-wall, NACA0012 polar
- Ballast-tank sloshing study: master curve pinned by measured VOF fields, passive roll-damper reduction, and a run-time estimator calibrated to the machine's own measured runs